

AI Library Assistant Agent

Problem Solving and Problem Formulation

1. Problem Formulation

Element	Description
Initial State	The agent starts when a student needs help finding a book or library resource. The available books and their information are stored in the library system.
Actions	Search for a book; check book availability; provide book details; suggest related books; help the student find the required resource; and provide basic borrowing information.
Transition Model	The state changes according to the student's request. For example: student request → search books → check availability → select suitable book → provide information to the student.
Path Cost	The cost can be measured by the time taken to find a suitable book and the number of unnecessary search steps. The agent tries to find the required resource with fewer steps.
Goal Test	The goal is reached when the agent finds the required book or suitable resource and provides the correct information to the student.
State Space	Different possible states of the library system, such as searching for a book, checking availability, finding a suitable book, and providing the final result.
Search Space	All possible ways of searching the library collection, checking different books, and selecting a suitable resource for the student.
Agents	Multiple agents are involved, including the Library Assistant Agent and the student. The library staff can also provide or update book information.

2. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	The agent can access the information available in the library system, but it may not know exactly what the student needs until the student provides a request.
Determinism	Stochastic / Non-deterministic	The result may change because book availability can change when books are borrowed or returned.
Episodes	Sequential	One action can affect the next step. For example, the search result is used to check the availability of a book.
Change	Dynamic	Library information can change when new books are added or books are borrowed and returned.
Values	Discrete	Books, categories, authors, availability, and search results are represented as separate values.
Agents	Multi-Agent	The environment includes the Library Assistant Agent, students, and library staff, who have different roles.

3. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The Library Assistant Agent goes through different states while helping the student. It starts with a book request, searches the library collection, checks availability, selects a suitable book, and then provides the result.

Final Answer

The Library Assistant Agent is best classified as a Multi-state agent because it goes through several stages while helping a student. It starts with the student's request and then searches for a suitable book, checks its availability, and provides the required information. The environment is partially observable, stochastic, sequential, dynamic, and multi-agent. These properties explain how the Library Assistant works with students and library resources.